

TABLE 4.1 Performance Statistics for Magic Formula (Market Weight), Earnings Yield, Return on Capital, and S&P 500 (Total Return) (1974 to 2011)

	Magic Formula	Earnings Yield	Return on Capital	S&P 500 TR
CAGR (%)	13.94	15.95	10.37	10.46
Standard Deviation (%)	16.93	17.28	17.04	15.84
Downside Deviation (%)	12.02	11.88	11.35	11.16
Sharpe Ratio	0.55	0.64	0.35	0.37
Sortino Ratio (MAR = 5%)	0.80	0.96	0.56	0.56
Rolling 5-Year				
Outperformance (%)	—	15.11	84.38	80.10
Rolling 10-Year				
Outperformance (%)	—	11.28	89.91	96.44
Correlation	—	0.927	0.806	0.872

Data Source: Wesley Gray and Tobias Carlisle. *Quantitative Value: A Practitioner's Guide to Automating Intelligent Investment and Eliminating Behavioral Errors*. (Hoboken: Wiley Finance) 2012.

17.04 percent for *return on capital*. While the Magic Formula did seem to have a slightly lower standard deviation than the *earnings yield* or *return on capital* alone, if we limited the statistic to just downside deviation, which measures just the extent to which the portfolios deviate below the mean return, we found that the Magic Formula's downside deviation was slightly higher than the *earnings yield* or *return on capital* alone. The Magic Formula's downside deviation was 12.02 percent, against 11.88 percent for the *earnings yield* and 11.35 percent for *return on capital*, which means that the *earnings yield*'s higher standard deviation was tilted toward the upside. For the period tested, the earnings yield had the best volatility-adjusted return ratios, earning a Sharpe ratio of 0.64 against the Magic Formula's 0.55 and *return on capital*'s 0.35, and a Sortino ratio of 0.96 against the Magic Formula's 0.80 and *return on capital*'s 0.56. On every return or volatility measure, the *earnings yield* beat out the Magic Formula, which means that *return on capital* did not reduce volatility. Neither did *return on capital* improve the Magic Formula's consistency. While the returns to the *earnings yield* and the Magic Formula were closely correlated—as one would expect because one is a component of the other—on a rolling 5-year basis, the *earnings yield* beat out the Magic Formula 15.1 percent of the time, and about 11.3 percent of the time on a rolling 10-year basis. What does *return on capital* contribute to the Magic Formula? Startlingly, it seems to add little but poorer returns and elevated volatility.